Research Statement

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Political scientists often study phenomena that cannot be observed directly. For example, we use responses to hypothetical survey questions to infer actual behavior, we resort to aggregate election results to understand individual evaluations of politicians' performance in office, and we conduct randomized controlled trials in some places to determine if a policy is advisable in other places. Doing this credibly requires creativity in research design, as researchers must try to anticipate the challenges to inference even before conducting data analysis.

My research develops standards to navigate the tradeoffs that emerge when one considerx research design alternatives before data collection. I use tools from design-based causal inference and computational social science to identify practices and procedures that researchers can adopt to improve experiments and quasi-experiments at the pre-analysis stage. My current agenda focuses on the design of field and survey experiments, as well as the analysis of geographic spillovers.

This methodological agenda informs a substantive research program on the challenges to accountability and representation in developing countries. In this region, the efforts to improve the relationship between citizens and elected officials have a mixed record, and better research design can help us disentangle questions that currently seem intractable.

Navigating research design tradeoffs

Improving precision before conducting an experiment

The last decade has seen considerable improvement in research transparency and registration. More recently, advances in experimental design have developed tools to diagnose the properties of a research design before data collection. For example, one can think about bias, power, or target sample size under different hypothetical data generation processes.

Contrary to most of the work in statistics, econometrics, and political methodology, My agenda in this topic focuses primarily on optimizing statistical precision instead of minimizing bias, which implies preferring the research design feature that increases the probability if detecting a non-zero effect (if it exists) over producing an estimate as close as possible to the hypothetical truth.

I emphasize precision over unbiasedness for two reasons. First, researchers often face the choice between alternative experimental designs for which unbiased estimators are already documented. Two projects reflect this tradeoff.

In a manuscript under review, I revisit double list experiments. Social scientists use list experiments in surveys when respondents may not answer truthfully to sensitive questions. When their assumptions are met, list experiments reduce sensitivity biases from misreporting. However, they tend to produce estimates with high variance, which prevents researchers from improving upon direct questioning. Double list experiments promise to remedy this by implementing two parallel list experiments and then aggregating their results, which roughly halves the variance of standard list experiment estimates. However, their implementation brings the

question over whether the aggregation of the results of two different experiments yields a valid estimate. I propose a statistical test to assess the validity of estimates from double list experiments by leveraging variation in the order in which respondents receive the sensitive item, which provides researchers with a tool to apply this technique more widely.

In a working paper with Erin Rossiter (Notre Dame), we discuss the circumstances under which adopting a research design feature aimed at improving precision can instead hurt precision through sample loss. For example, block randomization often improves precision, but if this requires contacting participants multiple times to collect blocking covariates first and then post-treatment outcomes, it creates space for attrition that would not exist otherwise. Both standard and block randomized experiments have well-known unbiased estimators, but one must choose whether to block or not depending on the expected gains in precision. Through simulation and the reanalysis of a survey experiment on misinformation in social media, we illustrate how researchers can entertain this precision-retention tradeoff at the pre-analysis stage.

The second reason to focus on statistical precision is that sometimes a more precise yet biased estimator gives more informative answers than an unbiased estimator. Two other projects reflect this issue.

First, in work in progress with Jake Bowers (Illinois) and Christopher Grady (USAID), we discuss the circumstances under which researchers should prefer biased yet more precise estimates in the analysis of experimental data. Once again, an example of this trade-off comes from the analysis of block-randomized experiments. Block-randomization entails grouping observations in blocks or strata, conducting parallel experiments in each, and then calculating a single treatment effect with a weighted average. Previous work in statistics suggests that block-size weights lead to an unbiased estimator of the average treatment effect in this setting. However, because experiments are expensive to implement, we illustrate situations in which researchers may prefer to sacrifice unbiasedness in favor of more precise estimates, which would lead to the use of precision weights instead, the equivalent of using fixed effects in OLS regression. We extend this idea to other estimation techniques, including covariance adjustment and M-estimation. We illustrate these ideas through simulation and by revisiting the design of a messaging field experiment conducted by the Office of Evaluation Sciences.

Second, in a chapter with Christopher Grady (USAID) and Jim Kuklinski (Illinois) in *The SAGE Handbook of Research Methods in Political Science and International Relations*, we discuss the trend of increasingly more complex research designs in survey experimentation. For example, short vignettes are replaced with factorial experiments including many attributes. This increase in complexity allows researchers to isolate causal effects from the role of potential confounders, but raises concerns over whether the tasks that respondents face in survey experiments tap into the intended construct outside of the survey framework. In other words, we draw attention to the trade-off between simple designs that risk confounding bias and complex designs that yield unbiased estimates but risk ecological validity.

Geographic spillovers as a model selection task

I also maintain a research program on navigating research design trade-offs in the analysis of geographic spillovers in quasi-experiments. In causal inference, spillovers or interference happens when the treatment assignment of one unit affect the outcome of others. I focus on the cases in which the researcher believes that spillovers occur through geographical proximity. Current methods to study spillovers from a causal inference perspective assume the researcher observes all the relevant pathways through which treatments travel.

[CONTINUE HERE WE KNOW THE PATHWAY BUT NOT HOW UNITS CONNECT WITHIN IT]

My first research program, recognized with the best poster award at the 2019 Latin American Political Method-

ology meeting, focuses on the analysis of spillover or interference effects in settings where the researcher observes units' placement in a network, but not how they connect to each other. Current methods to study spillovers from a causal inference perspective assume that the researcher observes all the relevant networks through which treatments travel. This assumption has two parts. First, the researcher can identify the appropriate network(s) to model spillovers. Second, the researcher observes how units influence each other within the network. Researchers can satisfy this assumption by designing an experiment meant to capture spillovers, yet in most applications one can assume the first part, but not the second.

To overcome this difficulty, I propose a protocol that uses pairwise distances within a network as parameters to assist in the identification of plausible models of spillovers. This allows researchers to be explicit about the gaps between theory and model specification, and to use their preferred data-driven model selection criterion to choose the most appropriate model among alternatives. I illustrate this protocol's properties and usefulness in the context of supervised learning model selection algorithms via simulation and with an application to the study of voting registration irregularities in Ghana.

Challenges to accountability and representation in developing countries

The unintended electoral consequences of exposing corruption

Gender corruption stuff [better title]

The literature in political science and economics on electoral accountability highlights voters' adverse selection problem: They prefer to have good over bad elected officials but they can only infer an incumbent's type through observable outputs of their performance. Therefore, corrupt incumbents have incentives to hide their illicit activities and appear good. To fight corruption, governments, civil society organizations, and scholars devote considerable resources to implementing and evaluating anti-corruption interventions that attempt to bridge the information gap between voters and politicians' performance in office. However, the cumulative evidence suggests that these interventions rarely translate into voters sanctioning corrupt incumbents.

A recent explanation for the lack of electoral sanctions is that politicians undertake preemptive behavior to ward off the negative consequences of corruption. For example, previous research shows that political parties avoid nominating legislators investigated for corruption to reelection. On the other side of the coin, research also shows that local level elected officials switch away from parties investigated for corruption after securing reelection. In short, politicians avoid direct association with corruption.

I expand on this research in work funded by the Lemann Center for Brazilian studies at the University of Illinois. I show how revealed corruption drives politicians in nearby localities to undertake preemptive behavior to ward off a negative reaction from their constituents. Whereas previous literature has shown evidence that partisans might react strategically to anticipated sanctions when their political party is linked to corruption, I argue that this behavior is even more widespread and that politicians fear being caught in an electoral anti-corruption wave even when there is no evidence of their own wrong-doing. This occurs because officeholders expect their constituencies to receive the news of corruption in nearby localities and react with increased scrutiny on their performance.

I advance this argument with data from a long-running anti-corruption program in Brazil that randomly selects municipalities to audit their use of federal funds, combining text analysis and machine learning to create the most comprehensive dataset of corruption infractions to date. I show that mayors in municipalities exposed to

nearby corruption are more likely to seek reelection under a different party, with subsequent analyses showing that they do so in a pattern that suggests an attempt to secure more resources for reelection.

I continue this research agenda in joint work with Kelly Senters Piazza (US Air Force Academy). In a forthcoming book chapter prepared for the Interdisciplinary Corruption Research Network, we use my data on corruption infractions at the local level in Brazil to revisit previous findings on the gendered electoral consequences of corruption. Previous work in this topic uses public opinion data and survey experiments to show how revealing corruption increases support for female representation, yet also how female politicians face harsher sanctions for comparable levels of wrongdoing. We argue that these methods of inquiry do not allow us to translate their conclusions to realistic settings, which we overcome by focusing on the aforementioned anti-corruption program in Brazil. We show that revealing corruption increases support for female representation in the form of an increased proportion of female candidates running for mayor, yet this support does not translate into an increased proportion of women winning elections, which we attribute to incumbent mayors' incentives to counter the rise of female challengers.

Our experience working on the gendered electoral consequences of corruption led us to produce timely insights on the COVID-19 pandemic's implications for female political representation. In a viewpoint piece published in *World Development*, we review recent work on how exceptional events, such as corruption, conflict, and public health disasters, promote female political representation by raising questions around the ability of male leadership to handle crises, as well as priming voters about issues around which women are perceived to have policy expertise. Examining public opinion data and recent media coverage around the world, we argue for the presence of both mechanisms in the current global pandemic, implying a potential surge in female political representation, which in turn can help alleviate the gender inequalities exacerbated by the pandemic.